



Mobile phone data

Module 2 - Session 2.2.1

Master data: configuration, contextual and reference datasets

FLOWMINDER.ORG

NOMMON

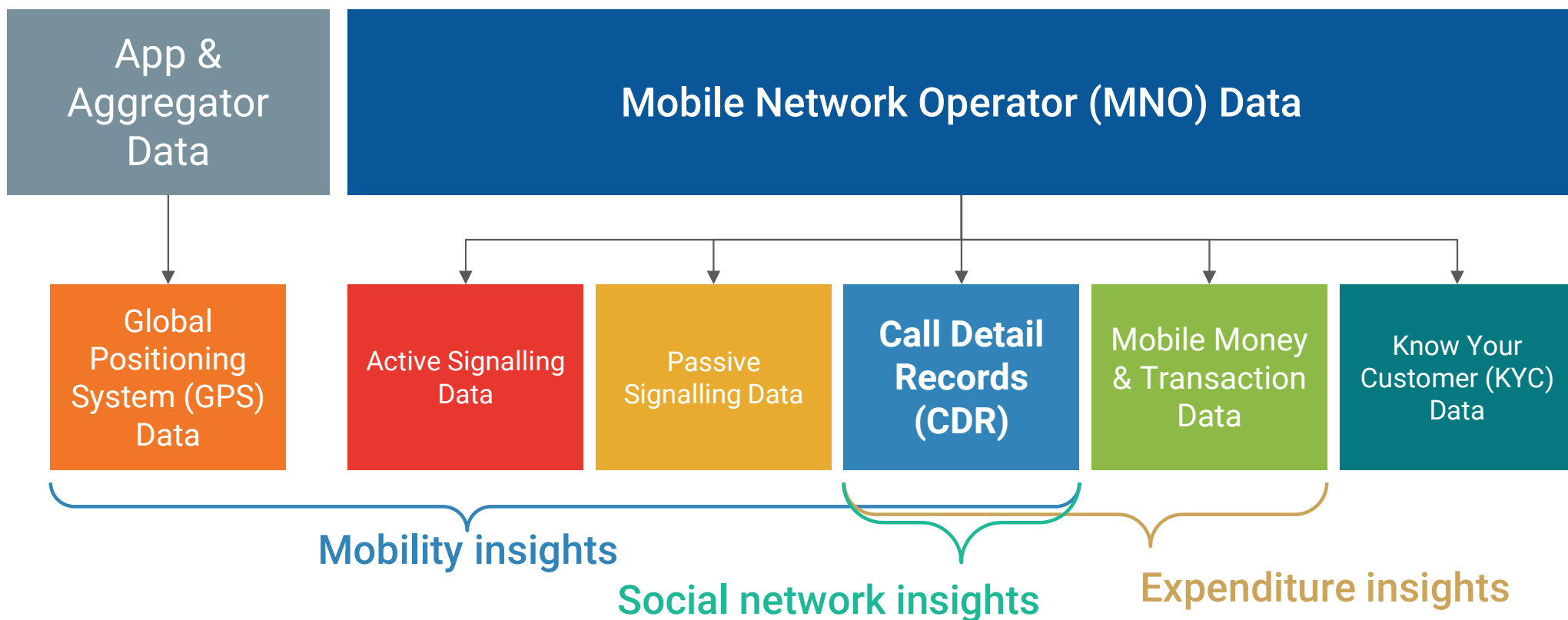
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Contents

1. Types of data produced by mobile phones and networks
2. Variables of mobile phone data for analysis
3. Strengths and limitations of mobile phone data
4. Technical characteristics of base transceiver stations (BTS) and radio cells
5. Issue of bias

Introduction to Mobile Phone Data

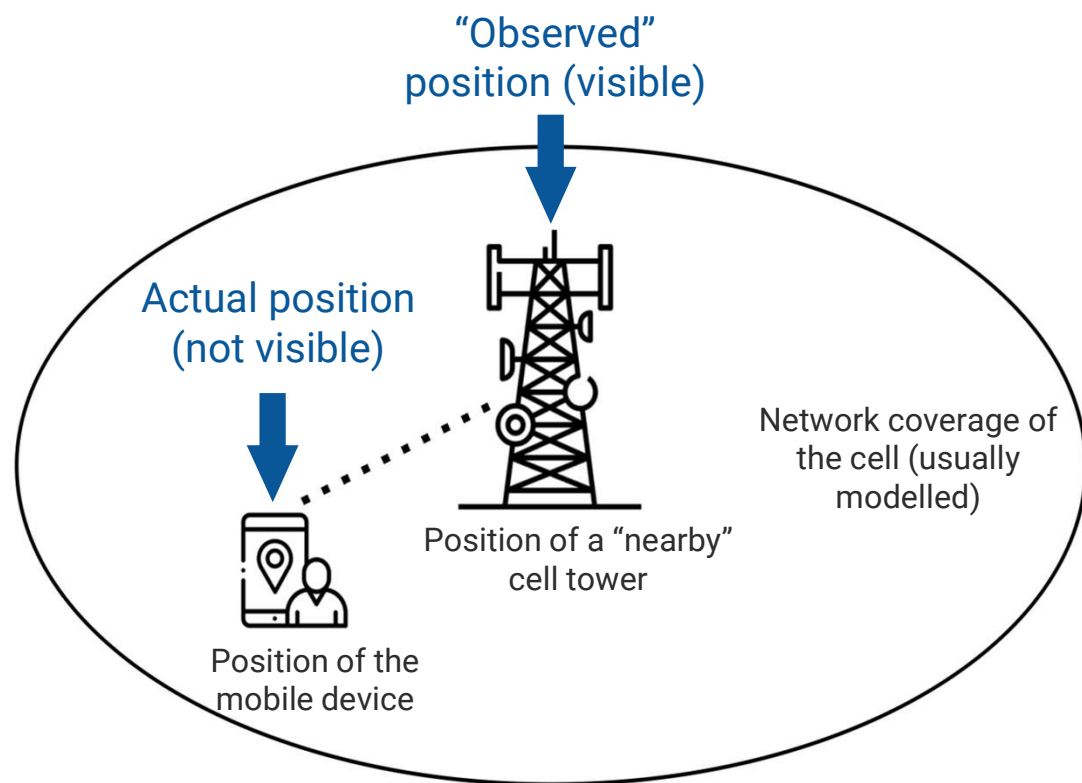
Mobile Phone Data (MPD)



How CDR data is generated

CDR data

- Passively generated when a subscriber
 - Makes or receives **a call**
 - Sends or receives **an SMS**
 - Uses mobile **data**
- Location is at **the cell tower level**
- Routinely stored by MNOs for billing purposes



MPD - GPS data vs CDR data

	Data collected /stored by	Spatial precision & resolution	Temporal resolution	Population coverage
CDR data	Mobile Network Operator	Varies across areas Approximate Cell tower location	Intermittent Depend on user behavior	All phones
GPS data	Smartphone app	Accurate Exact Device location	Continuous Frequent & constant	App/OS users of smartphones

Key information contained in CDR data

MSISDN	CELL_ID	TIMESTAMP
AA204V1542DCA00	451154211	2016-10-10 15:35:25
AA204V1542DCA01	451154211	2016-10-10 20:03:45
AA204V1542DCA02	451154211	2016-10-10 21:56:15
AA204V1542DCA03	451354312	2016-10-10 21:59:32
AA204V1542DCA04	452616792	2016-10-10 22:42:23
B45QHV45CAEVA5	476126941	2016-10-10 08:13:21
B45QHV45CAEVA6	476126941	2016-10-10 08:14:15
B45QHV45CAEVA7	476126941	2016-10-10 08:14:59
B45QHV45CAEVA8	476126941	2016-10-10 12:41:01
B45QHV45CAEVA9	476126941	2016-10-10 13:10:45
B45QHV45CAEVA10	476126941	2016-10-10 15:20:43

Calling party identifier
(pseudonymised)

Timestamp

Cell ID
(location)

Other variables in CDR data

- Receiving party identifier (pseudonymised)
- Network event type

CDR data do NOT contain information about the contents of a network event

Strengths and limitations

Strengths

- High penetration globally
- Large geographic scale
- High quality data
- Near-real time
- High granularity
- Automatically generated
- Already generated by organisations

Limitations

- Precision limited by cell tower density
- Representativeness of mobile phone subscriptions
- One subscription may not represent one individual, and vice versa
- Large datasets requiring suitable data infrastructure

Device location is observed as cell tower location

Time

6.00

6.35

7.25

8.00

9.40

10.45

13.00

14.50

16.00

17.30

19.00

20.20

20:50

21.35

22.50

23.45

Cell Tower

A

A

B

B

C

C

C

C

C

C

C

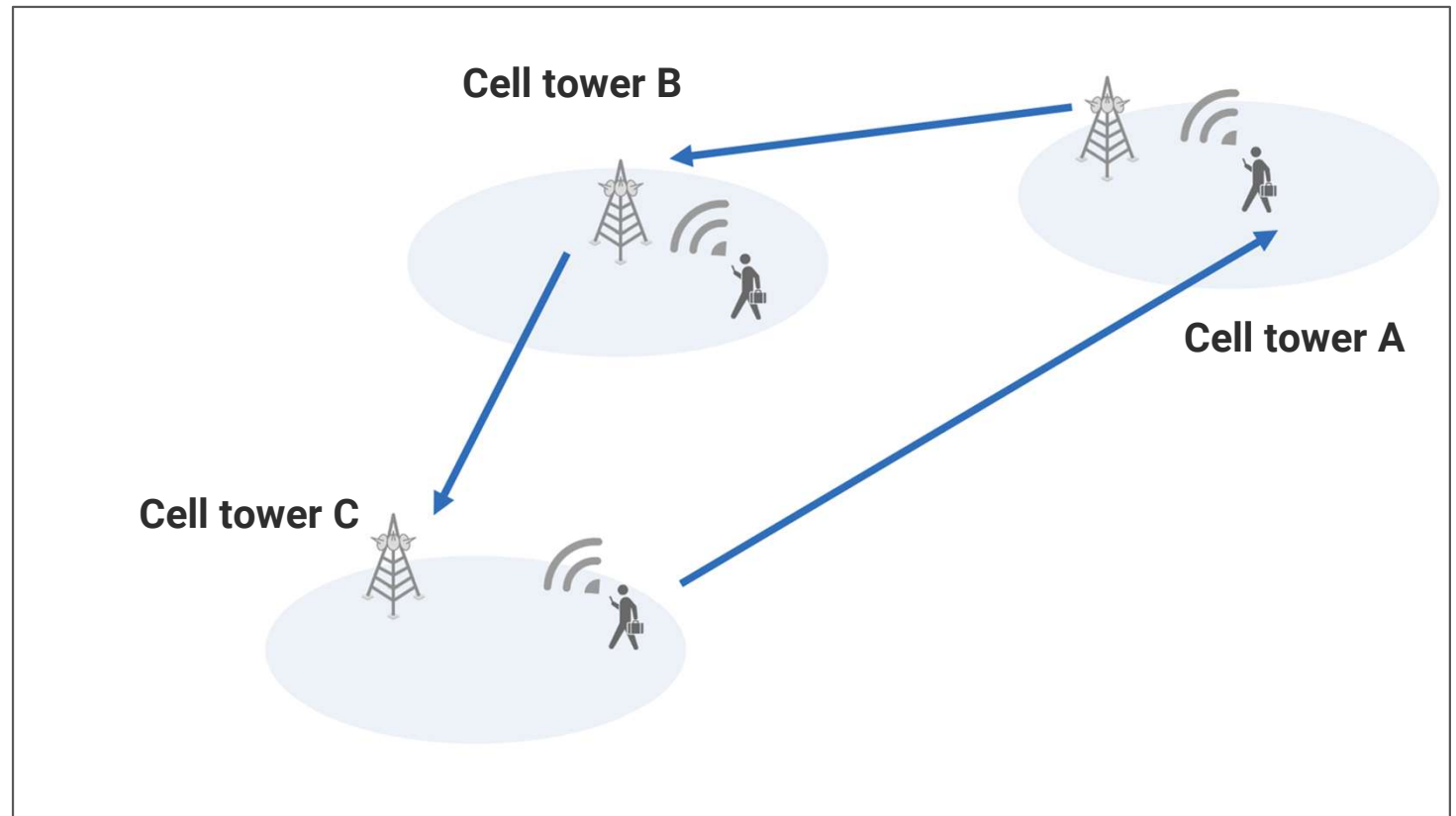
A

A

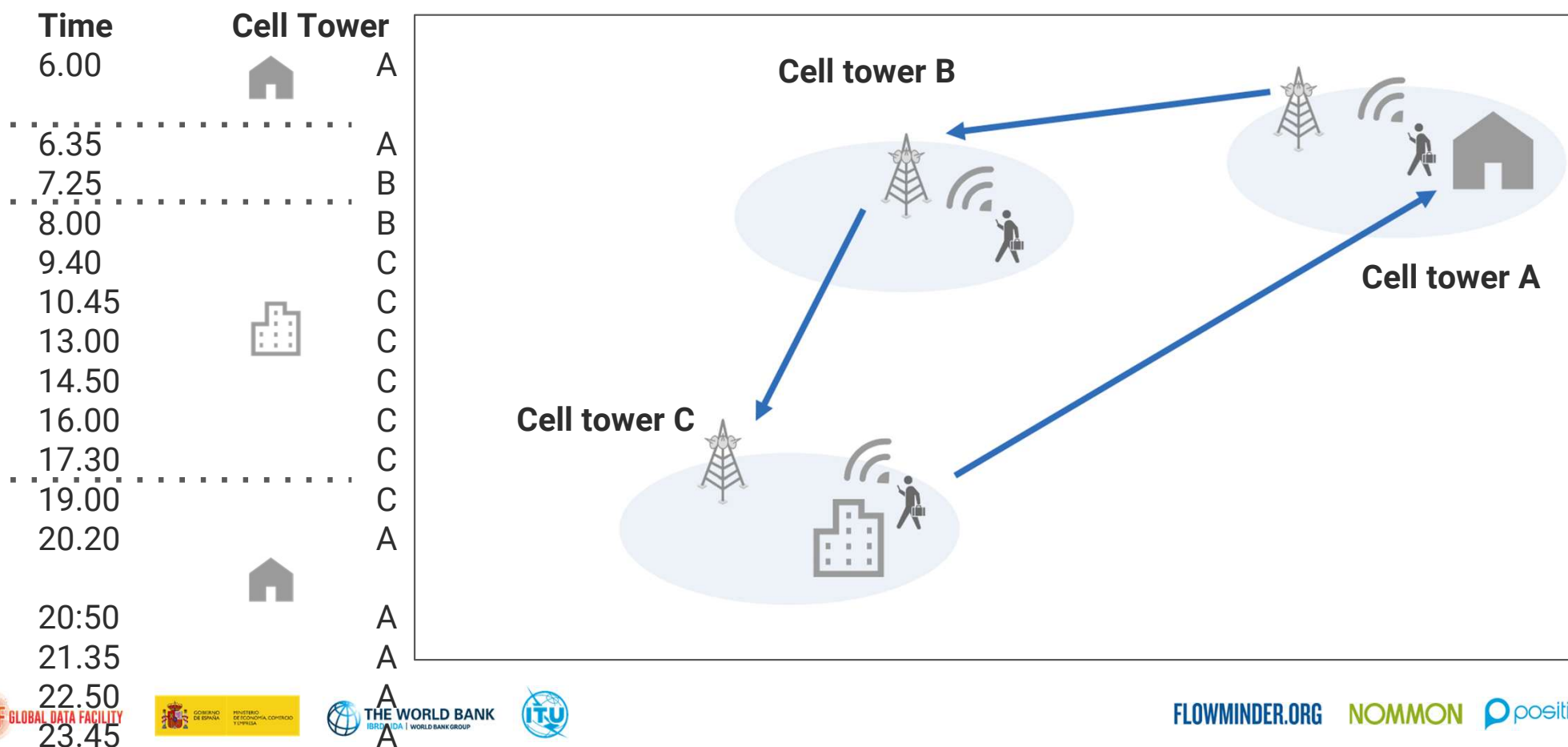
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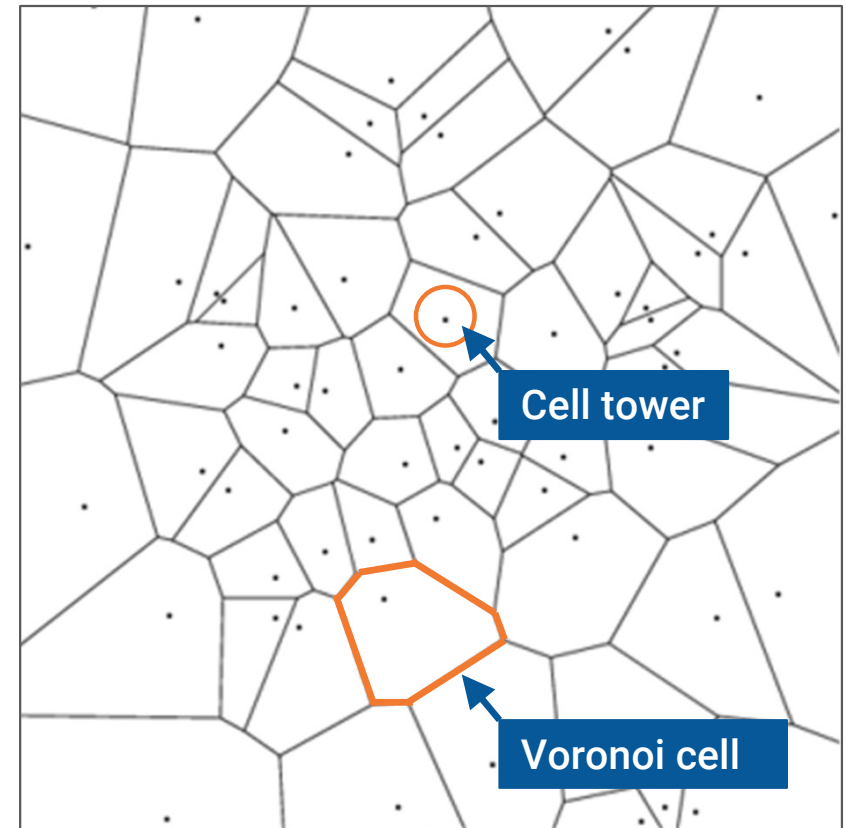


Timestamps are used to infer the type of locations

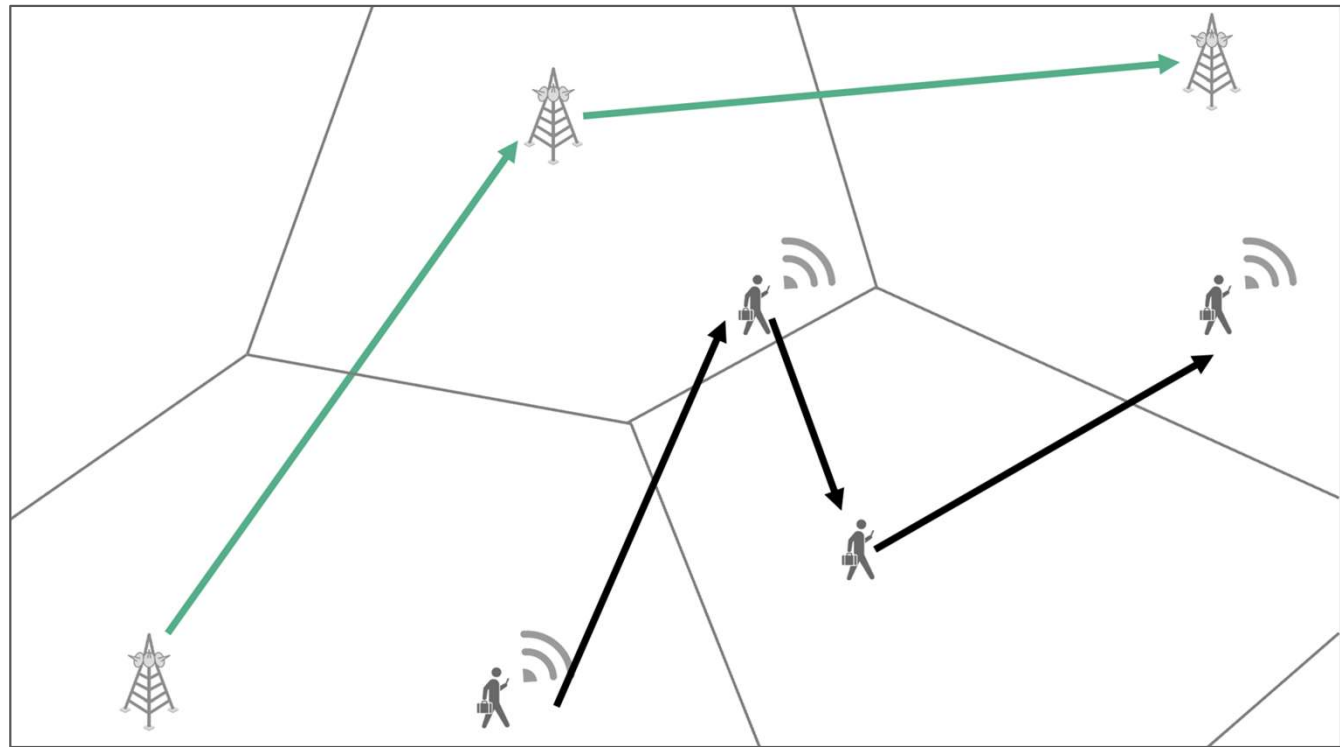
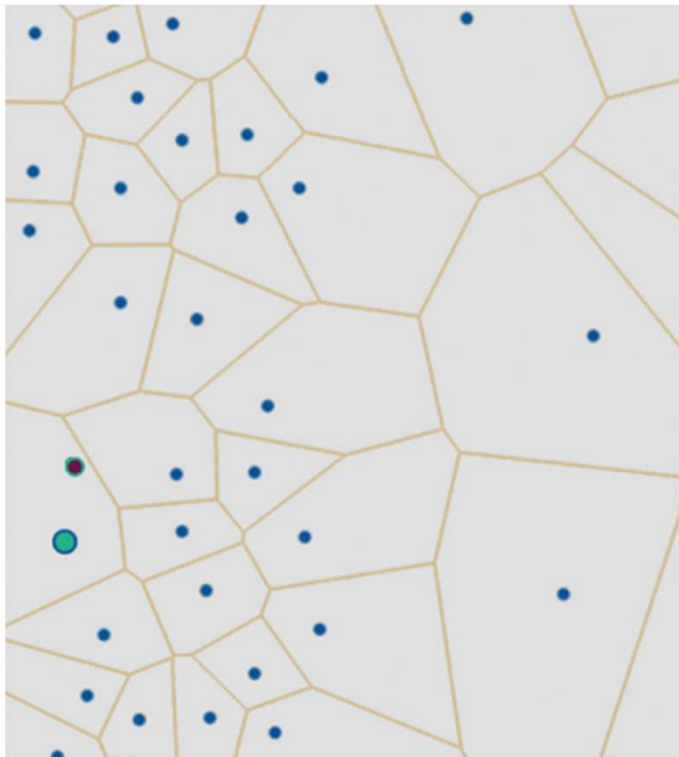


Hypothetical area represented by each cell tower: Voronoi cells

- A Voronoi diagram based on cell tower locations **divides a given area into regions associated with the nearest cell tower**
- Each region (voronoi cell) in the diagram **represents the hypothetical area that a cell tower network covers**



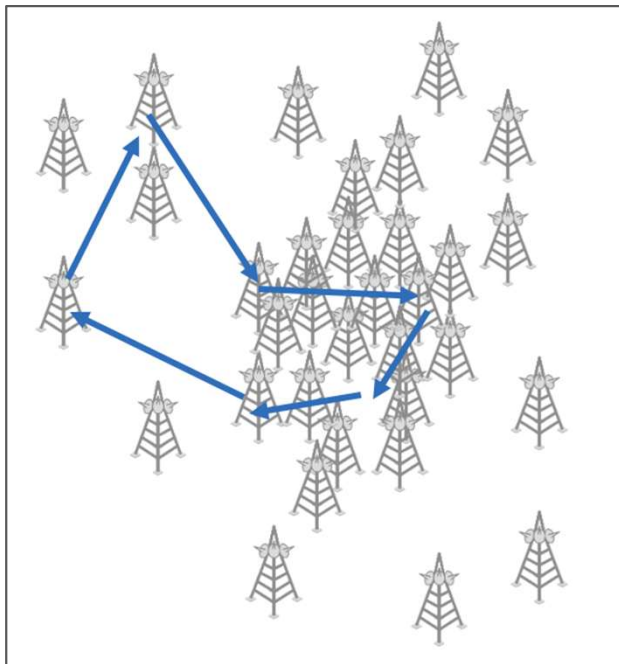
Flows among cell towers can deviate from actual mobility



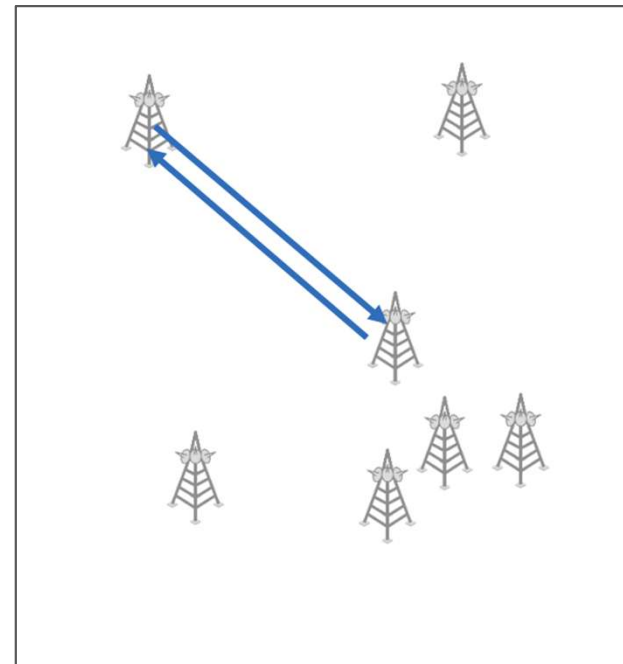
Subscriber movements Network events Observed trajectory
Cell tower coverage Cell towers

Cell tower density limit the mobility observable depending on localities

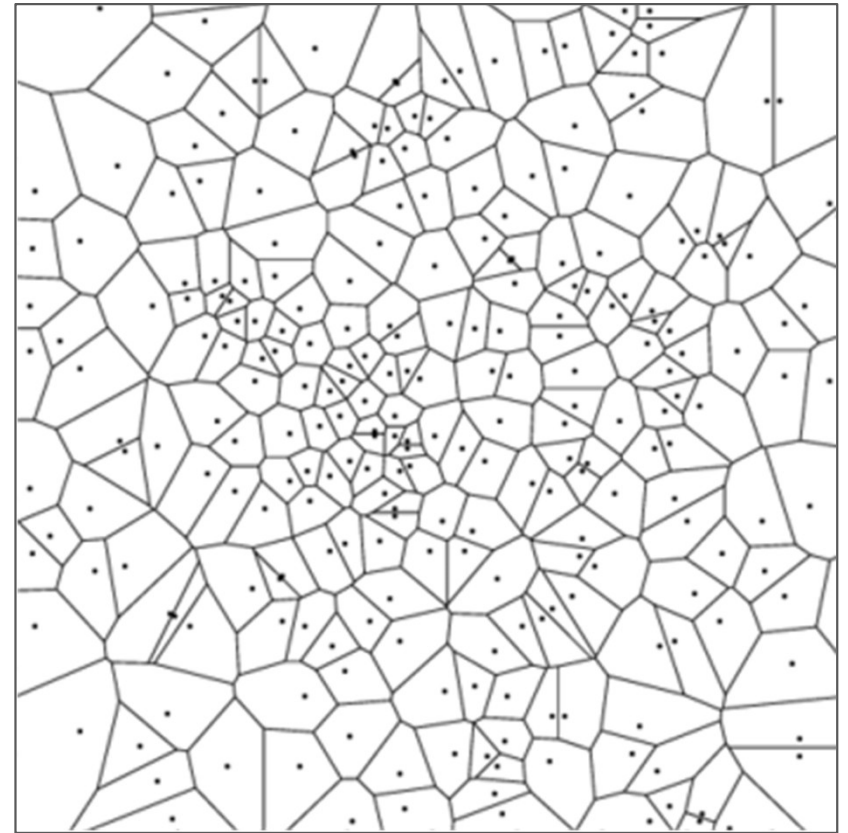
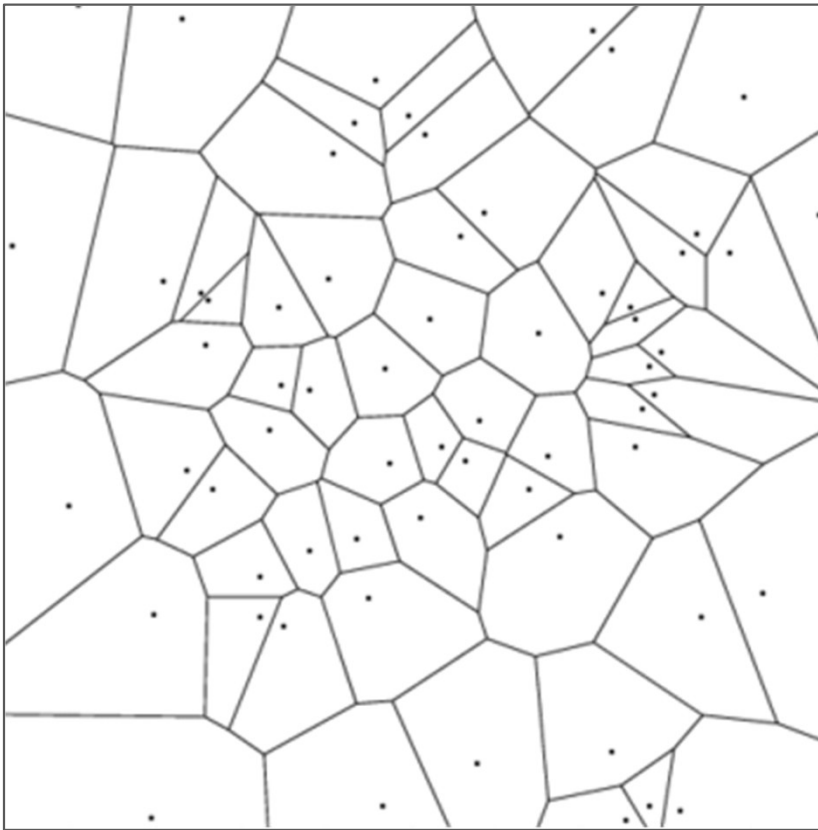
Urban



Rural

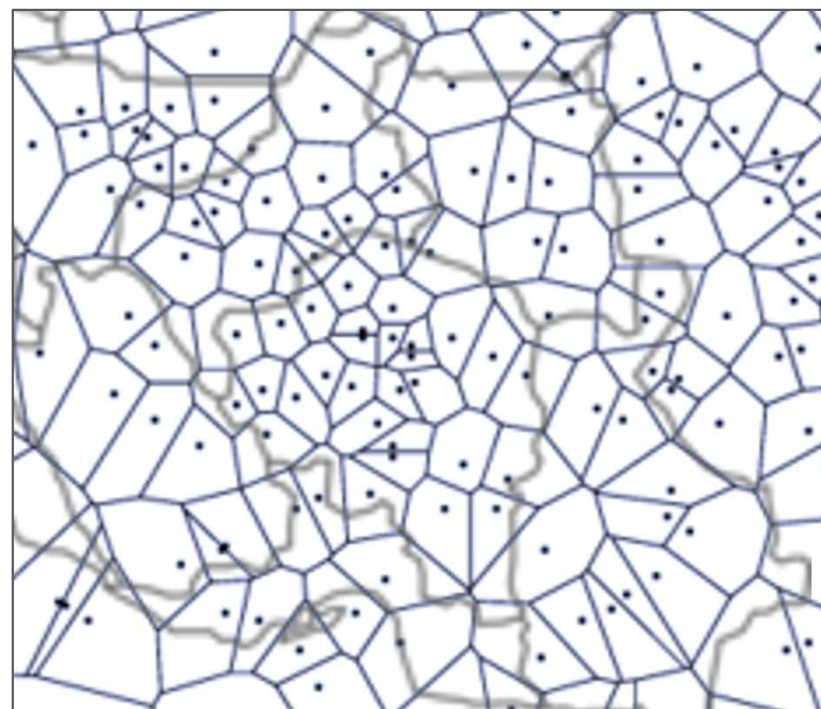


More cell towers allows for greater spatial precision



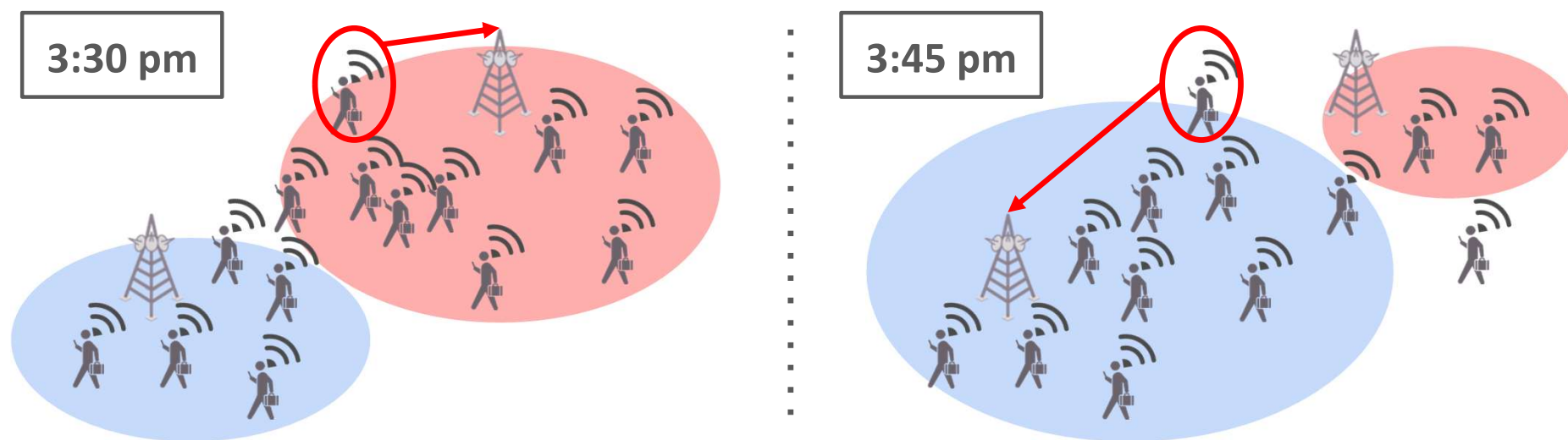
How can we convert population estimates at Voronoi cell to admin map?

- Residential population estimated for each Voronoi cell
- Overlay the voronoi map (—: blue) with the admin map(—:gray)
- If a Voronoi cell covers more than one admin units, proportionally allocate the estimated population according to the sizes of residential area or those of overlaps



Frequent tower switching (handover) causes location noise

- Tower switching occurs due to signal strength changes, load balancing, and technology switch by tower.

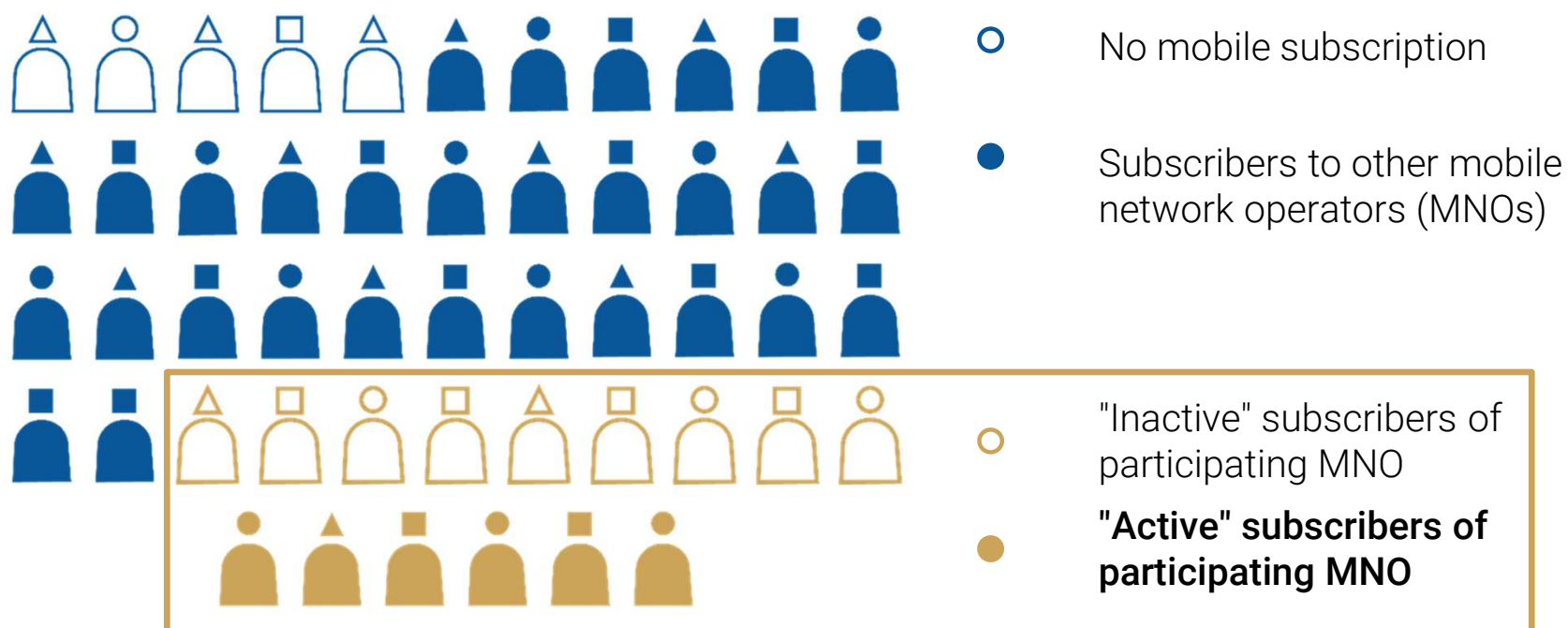


Usage habits affect the estimation accuracy

	Time Tower	Cell		Time Tower	Cell		Time Tower	Cell
	6.00	A		6.00	A		6.00	A
	6.35	A		6.35	A		6.35	A
	7.25	B		7.25	B		20.20	A
	8.00	B		8.00	B		20:50	A
	9.40	C		9.40	C		21:35	A
	10.45	C		17:30	C		22:30	A
	13.00	C		20:20	A			
	14.50	C		22:50				

Usage habits can vary by...
 Wealthier/poorer, age, gender, occupation, urban/rural (connectivity to the network), multiple-sim holding, phone sharing, etc.

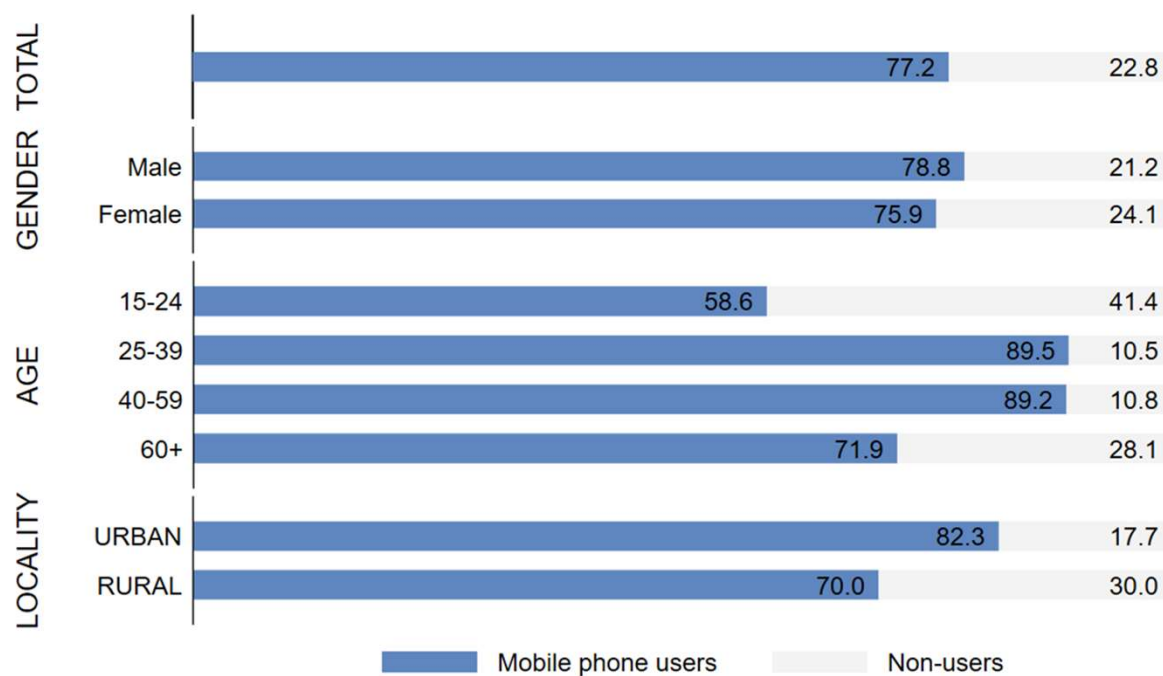
Who is in the CDR data you are accessing?



Mobile phone use by demographic | Ghana

Mobile phone use is:

- More common among **men** than among women
- More common among **persons of working-age** than among the elderly, and among children
- More widespread in **urban areas** than in rural areas



Source: AHIES 2022, Q3, weighted(pop_weight)

Issue of bias



A whole population
you are trying to estimate

Method for population-weighted adjustment

Data

CDR data, Census data (or existing population estimates such as gridded datasets), additional primary/secondary survey data

Method

1. Get initial estimate of the residential population for each cell tower and aggregate at the district level using CDR data
2. Compute the scaling factor for each district by dividing the census population by the CDR estimated population
3. Adjust the CDR-derived estimates by multiplying them by the scaling factors

Make further adjustment based on additional data availability (phone users & non-users, MNO subscription, multiple-SIM holding)

Examples of additional data for adjustment

Census: Gambia Bureau of Statistics (GBoS) included questions about ICT facility and device in the last Census conducted in 2023/24 (supported by the World Bank)

- Access to the mobile device
 - Mobile phone users or non-users
- Usage of the mobile services
 - Multiple-sim holding, phone sharing
 - MNO subscriptions

Computing scaling factor - The Gambia example

Scaling factor (SF) for the administrative unit (District i):

$$\text{SF}_i = \text{Resident}_{\{i, \text{allpop-census}\}} / \text{Resident}_{\{i, \text{subscribers-census}\}}$$

where

$\text{Resident}_{\{i, \text{allpop-census}\}}$: Total residential population of District i [Census]

$\text{Resident}_{\{i, \text{subscribers-census}\}}$: Number of subscribers residing in District i [Census]

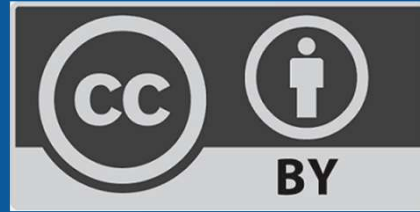
$\text{Resident}_{\{i, \text{subscribers-CDR}\}}$: Number of subscriber estimated to reside in District i [CDR]

$$\begin{aligned} &\text{Adjusted residential population estimate} \\ &= \text{SF}_i \times \text{Resident}_{\{i, \text{subscribers-CDR}\}} \end{aligned}$$



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Mobile Phone Data for Policy Programme (WB-GDF)
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